

L11: Deep Learning for Text

CSci 560 Deep Learning w/ Python (Chollet) Ch. 11

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L11.1 Natural Language Processing: The bird's eye view

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Natural vs. Computer Languages

- In computer science, we refer to human languages, like English or Mandarin, as **natural** languages, to distinguish them from languages designed for machines.
- Every machine language was **designed**: its starting point was a human engineer writing down a set of formal rules to describe the language.
 - Rules came first.
- Natural languages are shaped by an evolutionary process, much like biological organisms, that's what makes them *natural*.
 - Their “rules”, like the grammar of English, were formalized after the fact and are often ignored or broken by its users.

Conclusion: While machine-readable languages are highly structured and rigorous, natural language is messy - ambiguous, chaotic, sprawling, and constantly in flux.

Modern Machine Learning NLP Systems

Using machine learning and large datasets to give computers the ability to *understand* language.

Some examples of types of text processing systems:

- “What’s the topic of this text?” (text classification)
- “Does this text contain abuse?” (content filtering)
- “Does this text sound positive or negative?” (sentiment analysis)
- “What should be the next word in this incomplete sentence?” (language modeling)
- “How would you say this in German?” (translation)
- “How would you summarize this article in one paragraph?” (summarization)

In much the same way that computer vision is pattern recognition applied to pixels, NLP is pattern recognition applied to words, sentences, and paragraphs.

L11.2 Preparing Text Data

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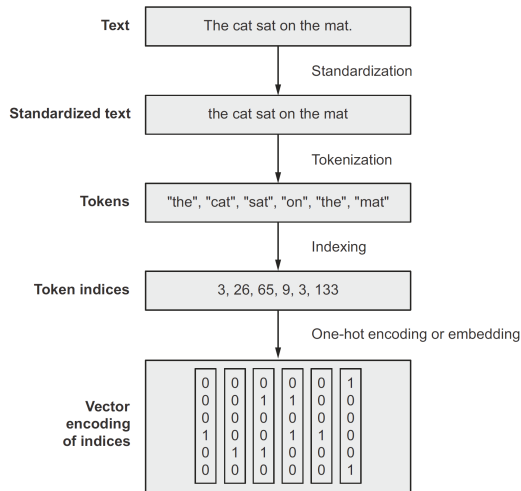
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Text Vectorization Process

Vectorizing text is the process of transforming text into numeric tensors.



- First, you standardize the text to make it easier to process, such as by converting it to lowercase or removing punctuation.
- You split the text into units (called tokens), such as characters, words, or groups of words. This is called tokenization.
- You convert each such token into a numerical vector. This will usually involve first indexing all tokens present in the data.

L11.3 Two Approaches for Representing Groups of Words: sets and sequences

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L11.4 The Transformer Architecture

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L11.5 Beyond Text Classification: Sequence-to-sequence learning

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Chollet, F. (2021). *Deep learning with python* (second). Manning.